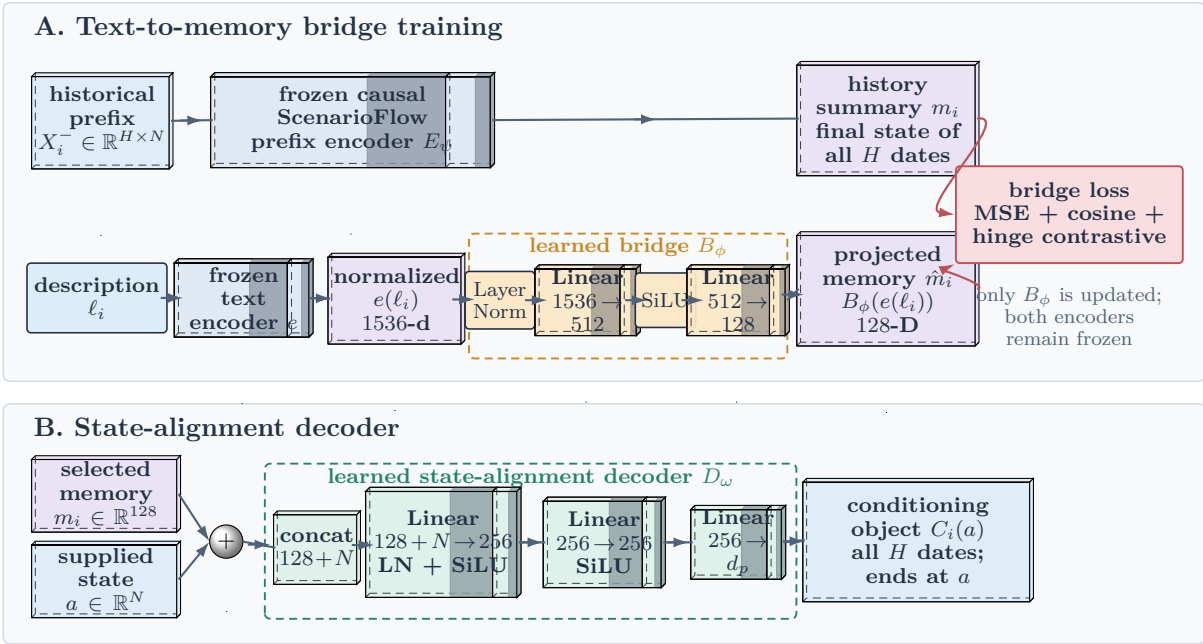

A. Text-to-memory bridge training
historical prefix $X_i^- \in \mathbb{R}^{H \times N}$
frozen causal ScenarioFlow prefix encoder E_ϕ
history summary m_i final state of all H dates
bridge loss MSE + cosine + hinge contrastive
description ℓ_i
frozen text encoder e
normalized $e(\ell_i)$ 1536-d
learned bridge B_ϕ
Layer Norm
Linear 1536 → 512
SiLU
Linear 512 → 128
projected memory $\hat{m}_i$ $B_\phi(e(\ell_i))$ 128-D
only B_ϕ is updated; both encoders remain frozen
B. State-alignment decoder
selected memory $m_i \in \mathbb{R}^{128}$
supplied state $a \in \mathbb{R}^N$
+
learned state-alignment decoder D_ω
concat $128+N$
Linear $128+N \to 256$ LN + SiLU
Linear $256 \to 256$ SiLU
Linear $256 \to d_p$
conditioning object $C_i(a)$ all H dates; ends at a